

The Riemann Hypothesis: Spectral Routes and Dead Ends

Part III of a Five-Part Series

Lukas Geiger

Bernau, Germany

ORCID: [0009-0005-7296-1534](https://orcid.org/0009-0005-7296-1534)

June 11, 2026

Abstract

This paper, Part III of a five-part series [8, 7, 6, 5], documents the spectral sector of the proof landscape for the Riemann Hypothesis. Two independent conditional reductions are presented:

Branch B (Hadamard Strip / I-1 Normal-Family Reframe) reduces RH to a single open hypothesis (ii-a): a uniform growth-adapted strip bound on the family $\{\hat{\xi}_\lambda\}$ derived from QW_N -coercivity. The multiplicative canonical form $A_\lambda(\delta) = A_\Xi(\delta)(1 + \kappa(\lambda)\delta^2)$ with $|\kappa_\infty| \lesssim 10^{-4}$ (corrected from the preliminary $[1.6, 1.8] \times 10^{-3}$; four-point exponential fit, $\lambda = 11$, $N \in \{137, 160, 200, 220\}$) yields the conditional strip bound $A_\lambda(\delta) \leq 1.0063$ under uniform log-curvature control.

Branch Z (CCM Microcluster Programme) reduces RH to three C2ca-inputs (scalar secular cancellation, projected Poisson quasimode, coercive complement) via the Connes–Consani–Moscovici Fourier model [2].

Both branches share Wall (ii) as the common analytical obstacle. Nineteen excluded spectral and CCM paths are catalogued, establishing the productive routes as the only viable approaches in the current landscape.

Series status (v3.0, 2026-05-26). The series is restructured from a trilogy to a five-part series; the conditional reduction status from v2.2 is unchanged. New in this part: the multiplicative canonical form (Section 2.4), Theorem B-1 (conditional uniform strip bound with $A_\lambda \leq 1.0063$, Theorem 2.3), and the RH-neutral Hadamard constant $c_\Xi = -\frac{1}{2} \sum_\rho 1/(\rho - \frac{1}{2})^2 \approx 0.0231$ (Section 4).

MSC 2020: 11M26, 11M36, 47A75, 30D15

Keywords: Riemann Hypothesis, spectral routes, normal family, Hadamard strip bound, CCM microcluster, conditional reduction, excluded paths

Contents

1	Introduction: The Spectral Sector	3
2	Branch B: Hadamard Strip / I-1 Normal-Family	3
2.1	The Reduction Chain (B1–B11)	4
2.2	The I-1 Normal-Family Reframe and the Foundation-Lock	4
2.3	Empirical Convergence (Mac Studio, converged dps)	4
2.4	Multiplicative Canonical Form	5
2.5	Theorem B-1: Conditional Uniform Strip Bound	6
3	Branch Z: CCM Microcluster Programme	6
3.1	The Reduction Chain (B1–B11, CCM-Branch)	6
3.2	The Three C2ca-Inputs	7
3.3	Empirical Convergence and the Effective Microcluster	7
3.4	The Determinant Channel: Zero-Counting Reformulation and Normalisation Ledger	8
4	Spectral Convergence Constants	9
4.1	Hadamard Curvature Constant c_{Ξ} (RH-neutral)	9
4.2	Residual Curvature κ_{∞} (Branch B)	9
4.3	Hecke-Normalised Gap $g_0(m)$ (Branch Z)	9
5	Common Wall and Spectral-Sector Outlook	10
6	Excluded Spectral Paths	10
7	Excluded CCM Paths (Branch Z Heritage)	11
8	Methodical Dead Ends	13
9	Summary	14

1 Introduction: The Spectral Sector

This paper covers the spectral sector of an ongoing multi-route investigation of the Riemann Hypothesis:

Part I [8]: Foundations and Obstructions — structural results (R1–R9), four critical obstructions (K1–K4), reorientation to Connes’ spectral program.

Part II [7]: Even Dominance (Branch A) — Shift Parity Lemma, computer-assisted finite-range diagnostics, Euler–Maclaurin Proposition, GF1 boundary-moment benchmark.

Part III (this paper): Spectral Routes (Branches B and Z) — Hadamard strip / I-1 normal-family reframe, CCM microcluster programme, excluded spectral and CCM paths.

Part IV [6]: Cross-Route Synthesis — combined analysis, dormant routes, external lines, methodical dead ends.

Part V [5]: Conclusio — condensed atlas, open problems, assessment and outlook.

The spectral sector attacks Wall (ii) — the approximation quality condition $k_\Lambda \approx \xi_\Lambda$ from Connes’ §6.6(b) [1] — through two structurally distinct mechanisms. Branch B decomposes (ii) via an I-1 normal-family argument into the strip bound (ii-a) and the limit-identification (ii-c), then absorbs (ii-c) into (ii-a) under a zero-simplicity assumption. Branch Z attacks (ii) through a microcluster quasimode-to-projector estimate in the CCM Fourier model. The two branches are independent of Branch A’s Asymptotic Variational Gap Conjecture (A7) from Part II [7].

Both branches share Wall (ii) with Branch A but attack it through fundamentally different analytical tools. The cross-route synthesis (patterns common to all three branches) is deferred to Part IV [6].

2 Branch B: Hadamard Strip / I-1 Normal-Family

Branch B is a parallel reduction to RH, independent of the Asymptotic Variational Gap Conjecture (A7) that conditions Branch A. Its starting point is the same as Branch A’s: Connes–van Suijlekom Theorem 5.6 [3] provides the finite- N error bound $\text{err}_\lambda \leq Ce^{-c\lambda}$ for each individual Riemann zero, verified numerically at $\lesssim 10^{-25\lambda}$ for ≥ 15 zeros. Branch B then reframes the limit condition via err-collapse, the I-1 Normal-Family decomposition, and the empirical data summarised below.

2.1 The Reduction Chain (B1–B11)

Step	Statement	Status	Source
B1	Connes' Theorem 6.1 / even dominance $\Rightarrow$ zeros of $\hat{\eta}$ real (= A1)	proven	[3, 1]
B2	Condition (i): real zeros of $\hat{\xi}_\lambda$ at finite N (CvS Thm 5.6 / 1-dim even kernel)	proven at every finite N	companion proof notes
B3	err-collapse equivalence: $\text{err} \rightarrow 0 \iff$ (i) + (ii)	rigorous reduction	companion proof notes §1
B4	err-collapse empirics: $\text{err} \sim 10^{-25\lambda}$ for ≥ 15 zeros, 4 λ -points	numerical evidence	companion proof notes §9
B5	I-1 Normal-Family Reframe: (ii) = (ii-a) $\wedge$ (ii-c) via Montel–Vitali compactness	rigorous decomposition	§E.2
B6	(ii-c) absorbed into (ii-a) via parity + Hadamard order-1 (under standing assumption (Z) on zero simplicity)	rigorous	§E.14
B7	Foundation-Lock: $\hat{\xi}(z) = (2/\sqrt{L}) \sin(zL/2)F(z)$ with $F(z) = \sum_{n \leq N} \Lambda(n)/\sqrt{n} \cdot \cos(z \log n/2)$; centered, even, real-on-real, entire	rigorous (algebraic identity + pole cancellation)	§E.11, §E.15
B8	Strip amplification flat at converged precision: $A_\lambda(0.4) \approx 1.004$ across $\lambda \in \{3, 5, 7, 9, 11, 13, 15\}$	numerical evidence, $\text{dps} \approx 30\text{--}50 \cdot \lambda$	§E.18, see Table 1
B9	(ii-a): uniform growth-adapted strip bound on $\{\hat{\xi}_\lambda\}$ from QW_N -coercivity	OPEN (the wall for Branch B)	§E.7, this paper
B10	MS2 closure (microcluster — three C2ca inputs)	conditional reduction (delegated to Branch Z)	[9]
B11	RH conditional on B9 $\wedge$ (delegated B10 via the err-collapse coupling)	conditional reduction	this paper + [4, 9]

2.2 The I-1 Normal-Family Reframe and the Foundation-Lock

The decomposition of (ii) into the strip bound (ii-a) and the limit-identification (ii-c) is the structural heart of Branch B. Montel–Vitali compactness on the normal family $\{\hat{\xi}_\lambda\}$ reduces local uniform convergence to two sub-conditions; the parity of QW_N (real-symmetric, even-parity projection $\Rightarrow \hat{\xi}_\lambda$ is even and real-on-real) together with a Hadamard order-one argument forces the Bohr–Carathéodory phase factor to be identically 1 under standing assumption (Z) (zeros simple). The canonical criterion object is then definitively fixed as the Foundation-Lock above (deviation from numerical data $\leq 1.9 \times 10^{-52}$ at $\lambda = 5, 7, \text{dps} = 50$).

Branch B therefore reduces *entirely to the single open condition (ii-a)*: a rigorous uniform growth-adapted strip bound derived from QW_N -coercivity. This is fewer open building blocks than Branch A's two (A3 + A7), and the empirical evidence for (ii-a) at converged precision is unusually strong (next subsection).

2.3 Empirical Convergence (Mac Studio, converged dps)

Converged data for the sup-observable $A_\lambda(\delta) := \sup_{x \in \mathbb{R}} |\hat{\xi}_\lambda(x + i\delta)|/|\hat{\xi}_\lambda(x)|$ and the L^2 -observable $B_\lambda(\delta)$ at $\delta = 0.4$, obtained on a dedicated compute node (Mac Studio, $\text{dps}_{\min} \approx 30\lambda\text{--}50\lambda$ to control cluster near-degeneracy; two distinct dps values bit-identical at $\lambda = 11, 13, 15$):

The near-flatness of $A_\lambda(0.4)$ shows that the family $\{\hat{\xi}_\lambda\}$ is strongly sub-generic, making (ii-a) empirically plausible. The L^2 -observable grows as $B_\lambda(0.4) \approx 0.68\lambda^{0.4}$ on the plateau, consistent with polynomial sub-generic behaviour. Earlier runs at $\text{dps} = 30$

λ	$\sup \hat{\xi}_\lambda _{\mathbb{R}}$	$A_\lambda(0.4)$	$B_\lambda(0.4)$	$B_\lambda(0.4)/\lambda^{0.4}$
3	0.858	1.0033	1.151	0.74
5	0.879	1.0037	1.340	0.70
7	0.886	1.0038	1.505	0.69
9	0.890	1.0039	1.649	0.68
11	0.890	1.0039	1.778	0.68
13	0.891	1.0039	1.895	0.68
15	0.892	1.00394	2.003	0.68

Table 1: Converged empirics for Branch B observables at $\delta = 0.4$ (companion proof notes, §E.18). The sup-observable $A_\lambda(0.4)$ is essentially flat at ≈ 1.004 , in striking contrast to the generic Phragmén–Lindelöf prediction $\sim \lambda^\delta$. The B_λ -ratio decreases from 0.74 at $\lambda = 3$ to a stable plateau at 0.68 for $\lambda \geq 9$.

(§E.16–E.17) were retracted due to under-converged eigenvectors; the corrected data are documented in §E.18 of the companion notes [4].

2.4 Multiplicative Canonical Form

The strip-amplification data (Table 1) admit a factored representation in terms of the completed Riemann ξ -function. Define the *Hadamard baseline*

$$A_\Xi(\delta) := \frac{\sup_{x \in \mathbb{R}} |\Xi(x + i\delta)|}{\sup_{x \in \mathbb{R}} |\Xi(x)|}, \quad (1)$$

where $\Xi(z) = \xi(\frac{1}{2} + iz)$ is the standard completed zeta function on the critical line. By the Hadamard product, $\log A_\Xi(\delta) \sim c_\Xi \delta^2$ for small δ , where the Hadamard curvature constant is (see Section 4)

$$c_\Xi = -\frac{1}{2} \sum_{\rho} \frac{1}{(\rho - \frac{1}{2})^2} \approx 0.0231, \quad (2)$$

summing over all non-trivial zeros ρ of $\zeta(s)$. This representation is *RH-neutral*: it does not assume the Riemann Hypothesis. Under RH, the sum specialises to $\sum_{k \geq 1} 1/\gamma_k^2$ with $\gamma_k = \Im(\rho_k)$.

The empirical data are well described by the *multiplicative canonical form*

$$A_\lambda(\delta) = A_\Xi(\delta) (1 + \kappa(\lambda) \delta^2), \quad (3)$$

where $\kappa(\lambda)$ is a residual curvature coefficient that saturates empirically:

$$\kappa_\infty := \lim_{\lambda \rightarrow \infty} \kappa(\lambda) \in [1.6, 1.8] \times 10^{-3} \quad (4)$$

(six λ -points including $\lambda = 20$, with $\kappa(20) = 1.527 \times 10^{-3}$ and stable deviation from the factored form). The saturation is consistent with the Foundation-Lock structure: $\hat{\xi}_\lambda$ converges to Ξ in a growth-controlled sense, and $\kappa(\lambda)$ measures the residual finite- λ strip-curvature excess.

Remark 2.1 (Pointwise identity, not uniform bound). Equation (3) is a pointwise algebraic decomposition at each δ , not a uniform strip bound. The sup over x is taken separately on each side. Converting this to a *uniform* $A_\lambda(\delta) \leq C$ for all λ and $\delta \in [0, \frac{1}{2})$ requires the separate log-curvature control of Theorem B-1 below.

2.5 Theorem B-1: Conditional Uniform Strip Bound

The following conditional theorem converts the empirical canonical form into a quantitative strip bound for Branch B.

Definition 2.2. For $\lambda \geq \lambda_0$ and $\delta \in [0, \frac{1}{2})$, define the log-curvature excess

$$H_\lambda(\delta) := \log \frac{A_\lambda(\delta)}{A_\Xi(\delta)}.$$

Theorem 2.3 (Conditional uniform strip bound). *Suppose there exist constants λ_0 and $K \leq 1.8 \times 10^{-3}$ such that $H_\lambda(\delta) \leq K\delta^2$ for all $\lambda \geq \lambda_0$ and $\delta \in [0, \frac{1}{2})$. Then*

$$A_\lambda(\delta) \leq B_\Xi \cdot \exp(K/4) \leq 1.0063, \quad (5)$$

where $B_\Xi := \xi(0)/\xi(\frac{1}{2}) = 1.005791795350375 \dots$ is the analytical upper bound for A_Ξ on $[0, \frac{1}{2})$.

Proof sketch. By the hypothesis, $A_\lambda(\delta) \leq A_\Xi(\delta) e^{K\delta^2}$. The Hadamard product for Ξ gives $A_\Xi(\delta) \leq B_\Xi$ for $\delta \in [0, \frac{1}{2})$, where $B_\Xi = \Xi(0)/\Xi(i/2) = \xi(0)/\xi(\frac{1}{2})$ is attained at $\delta = \frac{1}{2}$ (limit value). The exponential factor is maximised at $\delta = \frac{1}{2}$: $e^{K/4} \leq e^{1.8 \times 10^{-3}/4} \leq 1.00045$. Hence $A_\lambda(\delta) \leq 1.005792 \times 1.00045 \leq 1.0063$. $\square$

Remark 2.4 (Open proof slot). The hypothesis $H_\lambda(\delta) \leq K\delta^2$ is the *uniform log-curvature control*. Deriving it rigorously from the Foundation-Lock structure

$$\hat{\xi}(z) = \frac{2}{\sqrt{L}} \sin\left(\frac{zL}{2}\right) F(z)$$

is an isolated open proof slot for Branch B. The empirical data (Table 1) satisfy this condition with K in the $1.6\text{--}1.8 \times 10^{-3}$ saturation range, matching the conservative theorem bound.

3 Branch Z: CCM Microcluster Programme

Branch Z is the third independent attack on Wall (ii), running through the Connes–Consani–Moscovici Fourier model of the Weil quadratic form. It is companion to the standalone Zookeeper paper [9], which contains the full audit-trail of the microcluster analysis and the C2-lemma chain. This section documents the reduction-chain summary; the conditional reduction rests on three C2ca-inputs (scalar secular cancellation, projected Poisson quasimode, coercive complement) plus a fixed-waterline outer-gap assumption $g_* \geq g_0$ of order one.

3.1 The Reduction Chain (B1–B11, CCM-Branch)

The chain partly overlaps Branch B (shared external preprint inputs and the err-collapse coupling) but diverges in the closure mechanism (microcluster quasimode rather than I-1 strip bound). Labels B1–B11 are reused with the CCM-Branch reading.

Step	Statement (CCM-Branch reading)	Status	Source
B1	$W02_{\text{even}}$ is rank-one (pole at $s = 1$): $\tilde{C} = +4.864$, nullspace dim 30 at $(\lambda, N) = (3, 30)$	proven	cluster_mechanism_analysis.py
B2	PHP-eigenvalues = cluster-eigenvalues (agreement 10^{-14})	proven	cluster_mechanism.log
B3	Cluster eigenvectors q_k satisfy $F(\gamma) \approx 0$ at Riemann zeros	proven (10^{-14} to 10^{-17})	multi_eigenvec_riemann.log
B4	15-digit cancellation: pole-term + null-term ≈ 0	proven	weil_positivity_analysis.log
B5	$P_e(\gamma)$ orthogonal to cluster in nullspace (overlaps 10^{-21} to 10^{-10})	proven	weil_positivity_analysis.log
B6	Counterfactual: $F(z) \neq 0$ at non-zeros (fact or 10^{13} separation)	proven	weil_positivity_analysis.log
B7	Relative positivity $w_{\min}/\tilde{C} \rightarrow 0$ for $\lambda \rightarrow \infty$	proven (numerical)	lambda_positivity_convergence.log
B8	$w_{\min}$ converges at fixed N ; bounded asymptote ≈ -2.36	proven	lambda_positivity_convergence.log
B9	N-scaled convergence: $w_{\min}$ independent of N , $w_{\min}/\tilde{C} \rightarrow 0$	proven (partial), server run continues	n_scaled_convergence.log
B10	Resolvent identity: $F(\gamma) = \beta[\alpha(\gamma) - R(\gamma, w)]$ algebraic + numerically verified (10^{-13})	proven	b10_analytical_derivation.py
B11	MS2 closure: $k_\lambda \approx \xi_\lambda$ via three C2ca-inputs + fixed-waterline outer gap $g_* \geq g_0$	conditional reduction on C2ca.1 + C2ca.2 + C2ca.5 + g_* -assumption	[9], C2ca chain

3.2 The Three C2ca-Inputs

The Zookeeper paper [9] reduces MS2 (Connes’ §6.6(b) / err-collapse condition (ii)) to three named conditional lemmas (theorem-level subject to a separate H-spectral-gap assumption):

- **C2ca.1 Scalar Secular Cancellation:** algebraic secular identity proven; uniform decay $s_\lambda \rightarrow 0$ proven only under a separate Galerkin-truncation control. Numerical evidence (N -convergence-gated, 2026-06): $|\mu/\alpha| \approx 4\text{--}5 \times 10^{-7}$ across $\lambda = 3, 5, 7, 9$, stable in N (drifts $\leq 6\%$ up to $N=80$, digit-stable in precision) — i.e. a genuine finite Galerkin limit, *mildly increasing* in λ ($\approx 4.1 \rightarrow 4.9 \times 10^{-7}$). This supports *smallness*, not decay: the closure input $s_\lambda \rightarrow 0$ remains a purely analytical obligation.
- **C2ca.2 Projected Poisson Quasimode:** transfer formula $h = (1/n)P_0\Pi(E + B)$ proven; uniform analytical control of P_0 -leakage and boundary decay open. Numerical evidence: $p_\lambda \approx 2\text{--}3 \times 10^{-6}$.
- **C2ca.5 Coercive Complement:** finite Min–Max identity for a prescribed spectral window proven; the uniform order-one gap g_* is not derived from Cauchy-interlacing alone — it requires a separate H-spectral gap or fixed-waterline / slush-collar control (replaced by C2cf.1 *Fixed-Waterline Quasimode Projection*: in the finite self-adjoint Galerkin model, for any pre-chosen $g_0 > 0$, $\|(I - P_{\lambda, g_0})k\| \leq R_0/g_0$).

These three inputs plus the fixed-waterline assumption replace the earlier mass-adaptive gap rhetoric (which was diagnosed as circular against MS2, see Section 7).

3.3 Empirical Convergence and the Effective Microcluster

The mass-based effective cluster gap $g_{\text{eff}}(\varepsilon) := u_{J+1} - u_0$ with $J_\varepsilon := \min\{J : M(J) \geq 1 - \varepsilon\}$ stabilises at $g_{\text{eff}}(10^{-12}) \approx 5\text{--}6$ (order one, λ -independent), giving $R_0/g_{\text{eff}} \approx 5 \times 10^{-7}$. The earlier fixed-tolerance gap was unstable (cut inside the genuine cluster); the mass-based replacement is stable and macroscopic. See Zookeeper C2by/C2bz for the full analysis [9].

3.4 The Determinant Channel: Zero-Counting Reformulation and Normalisation Ledger

After the literal MS1 (uniformly simple cluster splitting) was excluded (Section 6), the working reformulation of MS1 is *zero counting*: compare the CCM secular function

$$F_N(z) = \sum_{j=-N}^N \frac{\xi_j}{z - p_j}, \quad p_j = \frac{2\pi j}{\log c},$$

(ξ = even ground state of the Galerkin matrix) against the completed Ξ via the argument principle on contours, instead of asking for eigenvalue simplicity. Writing $P(z) := F_N(z) \prod_j (z - p_j)$ for the degree- $2N$ secular polynomial, a dedicated measurement campaign (2026-06; scripts `ms1_det_zero_counting_harness.py`, `ms1_newton_power_sum_balance.py`, `ms1_cartwright_ledger.py`; full audit trail in the Zookeeper companion [9]) established the following RH-neutral structure ($c = 9, 14, 25$; $N \leq 48$; all statements numerical, no MS1/MS2 closure is claimed).

- **Root anatomy.** All $2N$ roots of P are real, in $\pm$ -pairs: K_h hit pairs matching γ_k to ≥ 30 decimal digits inside the comb range (earlier $\sim 10^{-15}$ reports were limited by hard-coded reference zeros, not by the roots), a graduated degradation governed by the comb reach $p_N = 2\pi N / \log c$ as the *only* empirically supported scale (a conjectured density-equality scale $T^* = 2\pi c$ left no signature in six dedicated runs), *no* comb-shadow roots, and a c -dependent but N -stable number of outer tail roots (5/7/10 pairs at $c = 9/14/25$).
- **Far field.** The tail roots carry the Newton power-sum defects exactly ($S_m(\text{tail})/c_m = 1.000$ for $m = 4, 6$ at all three c): the Hadamard envelope of Ξ is realised by P as a thinned outer node cloud (reminiscent of Gaussian-quadrature nodes; the analogy is used descriptively only).
- **Near field (exact ledger).** On a packet contour around $\{\gamma_1, \gamma_2\}$ the logarithm of the pairing ratio decomposes exactly into four blocks — hit mismatch, tail cloud, uncanceled zeros, comb — and the decomposition reconstructs every measured Rouché margin to 3–4 significant digits (35 configurations). The block hierarchy is $\text{comb} \gg \text{uncanceled zeros} \gg \text{tail} \gg \text{hit mismatch}$.
- **Canonical pairing.** Scanning all comb-compensation windows, the margin minimum sits at the *bare* pairing P vs. $C\Xi$ — no comb factor in the reference (margins 1.56/1.65/1.73 at $c = 9/14/25$, $N = 40$). Completing the comb to its closed form $\prod_{j \geq 1} (1 - z^2/p_j^2) = \text{sinc}(z \log c / 2)$ is the *worst* reference: its exponential type $(\log c)/2$ in $\text{Im } z$ clashes with the polynomial P — the Cartwright-type obstruction in pure form.
- **Local convergence rate.** The bare-pairing margin falls from 3.86 ($N = 16$) to 1.65 ($N = 40$) and is nearly c -independent at fixed N . Its source is almost exclusively the uncanceled zeros above the hit reach $T_{\text{hit}}(N)$ (≈ 50 at $N = 16$, ≈ 100 at $N = 40$); the measured spread ratio matches the prediction $\sigma_1(\gamma > T_{\text{hit}})$ from the Riemann–von Mangoldt density [11] within 1%.

Honest wall statement. The channel settles the normalisation question of the regularised-determinant formulation at finite N : the reference is the bare $C_N \Xi$,

and the local convergence $P_N \rightarrow C_N \Xi$ is quantified to first order with rate $\sigma_1(\gamma > T_{\text{hit}}(N)) \sim \log(T_{\text{hit}})/T_{\text{hit}}$. That $T_{\text{hit}}(N) \rightarrow \infty$ — the hyperconvergence of the secular roots onto the zeros — is exactly the unproven content of MS1; nothing in the ledger circumvents it. The open surface of the determinant channel is thereby concentrated in a single quantified rate.

4 Spectral Convergence Constants

Both spectral branches produce characteristic constants whose identification and saturation provide evidence for the conditional reductions. We collect them here for comparison; the cross-route synthesis with Branch A's constant $C^* = 64\pi^2/3$ (Part II [7], § GF1) is deferred to Part IV [6].

4.1 Hadamard Curvature Constant c_Ξ (RH-neutral)

The Hadamard product for the completed Riemann ξ -function gives

$$\log \frac{|\Xi(it)|}{|\Xi(0)|} = \sum_{\rho} \log \left| 1 - \frac{-t^2}{(\rho - \frac{1}{2})^2} \right| \sim -\frac{t^2}{2} \sum_{\rho} \frac{1}{(\rho - \frac{1}{2})^2} + O(t^4),$$

whence

$$c_\Xi = -\frac{1}{2} \sum_{\rho} \frac{1}{(\rho - \frac{1}{2})^2} \approx 0.0231. \quad (6)$$

This is the leading quadratic curvature of $\log A_\Xi(\delta)$ near $\delta = 0$. The representation (6) sums over *all* non-trivial zeros and is *RH-neutral*. Under RH, $\rho = \frac{1}{2} + i\gamma_k$ and the sum specialises to $\sum_{k \geq 1} 1/\gamma_k^2$.

4.2 Residual Curvature κ_∞ (Branch B)

The multiplicative canonical form (3) decomposes A_λ into the Hadamard baseline A_Ξ and a residual factor $1 + \kappa(\lambda)\delta^2$. Empirically, $\kappa(\lambda)$ was initially reported as saturating at $[1.6, 1.8] \times 10^{-3}$; however, a dedicated N -convergence study (four-point exponential fit at $\lambda = 11$, $N \in \{137, 160, 200, 220\}$) shows $|\kappa_\infty(11)| \lesssim 10^{-4}$, approximately $15\times$ smaller. The earlier values conflated transient $\kappa(N)$ with the asymptotic limit. The sign of κ_∞ remains open ($\kappa_\infty(3) \approx -2.3 \times 10^{-3}$, negative and flat). As $\lambda \rightarrow \infty$, $\hat{\xi}_\lambda \rightarrow \Xi$ in the relevant topology; the residual curvature measures the finite- λ correction. The load-bearing requirement for Branch B (ii-a) is $\sup_\lambda |\kappa(\lambda)| < \infty$ (boundedness), not a specific sign.

4.3 Hecke-Normalised Gap $g_0(m)$ (Branch Z)

For the m -th Riemann zero γ_m , the classical Montgomery–Odlyzko pair-correlation heuristic [10] suggests a local mean zero spacing of $2\pi/\log \gamma_m$. The effective microcluster gap g_{eff} of Branch Z (Section 3.3) is consistent with this scaling: $g_0(m) \approx 2\pi/\log \gamma_m$.

Branch	Constant	Value	Status
B	c_{Ξ} (Hadamard curvature)	≈ 0.0231	analytical (RH-neutral)
B	κ_{∞} (residual curvature)	$ \kappa_{\infty}(11) \lesssim 10^{-4}$	corrected (preliminary $[1.6, 1.8] \times 10^{-3}$ was transient)
B	$B_{\Xi} = \xi(0)/\xi(1/2)$	$1.00579\dots$	analytical
Z	$g_0(m) \approx 2\pi/\log \gamma_m$	varies	classical (Montgomery–Odlyzko)

5 Common Wall and Spectral-Sector Outlook

The three branches share Wall (ii) as the common analytical obstacle, but attack it through structurally different mechanisms. The following comparison table summarises the conditional reductions.

Branch	Reduction chain	Open hypothesis (the wall here)	Status
A — Even Dominance	A1 + A2 + A7 $\Rightarrow$ RH	A7 (Asymp. Variational Gap Conj.) + A3 (Odd-sector lower bound)	conditional reduction
B — I-1 Normal-Family	B1 + B5 + B9 $\Rightarrow$ RH	(ii-a) (uniform strip bound from QW_N -coercivity)	conditional reduction
Z — CCM Microcluster	B1 + C2ca + fixed-waterline $\Rightarrow$ RH	C2ca.1 + C2ca.2 + C2ca.5 + $g_* \geq g_0$	conditional reduction

Shared inputs. A1 = B1 (both rest on Connes–vS Thm 5.6 [3]). The Foundation-Lock $\hat{\xi}(z) = (2/\sqrt{L})\sin(zL/2)F(z)$ from Branch B is the canonical criterion object used in Branch Z as well.

Distinct mechanisms. Branch A attacks Wall (ii) via eigenvalue ordering $\lambda_1^+ < \lambda_1^-$ (Part II [7]). Branch B attacks via a strip-uniform Hurwitz-style normal-family argument (this paper). Branch Z attacks via a microcluster quasimode-to-projector estimate (this paper + Zookeeper [9]).

Closest to conditional closure. Branch B currently has the smallest open surface (a single named hypothesis (ii-a)) together with strongly sub-generic empirical evidence ($A_{\lambda}(0.4) \approx 1.004$ flat over seven converged λ -points, Theorem B-1 giving $A_{\lambda} \leq 1.0063$ under the log-curvature hypothesis). Branches A and Z remain conditional reductions in the multi-gap sense (two or three open inputs respectively). The title of the present series follows the conservative convention: “Atlas” applies as long as no single branch has reached conditional-proof level under the audit protocol of the project `DECISION.md`.

Promotion protocol and future scenarios are discussed in Part V (Section 9 gives a brief forward reference).

6 Excluded Spectral Paths

The post-v2.0 spectral programme (Connes–van Suijlekom reformulation, I-1 normal-family reframe, err-collapse diagnostic) closed several routes. These are documented here for the audit trail; they do not affect the main conditional reduction.

- **Eigenvalue-ordering via a uniform spectral gap.** The plan to lift $\lambda_{\min}(W^+ - W^-) < 0$ to the eigenvalue inequality $\lambda_1^+ < \lambda_1^-$ via Kato–Temple, Davis–Kahan, or Lehmann gap estimates fails because the relevant minimum eigenvector of QW_N sits in a near-degenerate cluster (even/odd split $\sim 10^{-65}$ at $\lambda = 5$, $\sim 10^{-84}$ at $\lambda = 7$, super-exponentially collapsing with λ). No uniform spectral gap is available.

- **Localization hypothesis for the strip bound (ii-a).** A natural plan was to bound $|\hat{\xi}_\lambda(x + i\delta)|$ via localization assuming ξ_λ is concentrated near $t = 0$. This is falsified: ξ_λ is genuinely edge-concentrated ($r_{99} \approx L/2$, $|\xi_\lambda(0)| \approx 0$). A growth-adapted normalisation Φ is required.

- **Edge-profile factorisation for (ii-a).** The exact identity

$$F_\lambda(z) = 2 \cos(zL/2) C_\lambda(z) + 2 \sin(zL/2) S_\lambda(z)$$

isolates the explicit λ^δ -growth modulation. The edge route yields no type gain because the eigenfunction mass is spread ($r_{50} \approx 1$ uniformly, but $r_{99} \approx L/2$).

- **Literal MS1 (cluster ladder).** The conservative form of MS1 — uniformly simple splitting of the minimum eigenvalue of QW_N at every (λ, N) — fails: the splitting collapses super-exponentially with λ . Only a cluster/quotient reformulation has structural traction.
- **Strip-amplification at $\text{dps} = 30$ (retracted).** Early measurements at $\text{dps} = 30$ suggested sub-generic growth and non-monotone instability. At converged precision ($\text{dps} \approx 30\text{--}50 \cdot \lambda$), the picture is qualitatively different: $A_\lambda(0.4) \approx 1.004$ is flat. The methodological lesson (dps requirement scales linearly with λ for cluster-near-degenerate eigenvectors) is itself substantive.
- **Raw- L^2 Hurwitz passage (falsified 2026-05-15).** The plan to derive the Hurwitz strip-uniform convergence directly from a raw- L^2 tail estimate fails on two fronts: (a) the archimedean tail has norm $O(\lambda^{-1/2})$ — exactly boundary exponent $\alpha = 0$, no margin; (b) the prime-Dirac tail diverges in the $m = 1$ branch. Negative-Sobolev does not rescue this. The diagnostic identifies operator-norm topology (i.e. Branch B) as the only viable replacement. The Connes–Hurwitz–Passage project re-scoped on 2026-05-17 to a purely expository role (Geiger2026hurwitz, in preparation).

Remark 6.1 (Common pattern). These six exclusions share a structural pattern: the minimum-eigenvector cluster of QW_N is the source of the technical difficulty. Methods that assume or require a uniform gap, a thin-localised eigenfunction, or a generic function-class behaviour all fail. Productive routes work *with* the cluster (waterline / cluster-quotient reformulations, growth-adapted normalisations) rather than against it.

7 Excluded CCM Paths (Branch Z Heritage)

The CCM microcluster programme accumulated its own catalogue of dead routes. They do not affect the Branch-Z conditional reduction.

- **Mass-based microcluster definition.** Defining the microcluster via a mass criterion is circular against MS2. Replaced in Zookeeper v1.2+ by a *spectral* microcluster (low-energy eigenspace bounded by an outer gap g_* of order one).
- **Naive Hurwitz passage.** A direct Hurwitz application without approximation-quality control yields statements too weak for the err-collapse closure. Replaced by a Slepian–Pollak-/PSWF-based passage.

- **Anchor-perturbation / Banach contraction (C2br', falsified).** A Banach-norm contraction $C_{\max} < M$ fails by factor 55–7000 on all twelve tested configurations.
- **Projected frontier cancellation $S_{bulk} \approx -S_{bd}$ (C2bs, falsified).** Both contributions have the same sign and are both of size $\sim 10^{-7}$. The true cancellation mechanism sits in the eigenvalue equation $(A - w_0)k \approx 0$.
- **Uniform mass criterion C2ak.** The total off-cluster mass bound diverges with λ ($6.68 \rightarrow 15.3 \rightarrow 1865$ across $\lambda = 3, 5, 7$) because it discards internal cancellation.
- **Canonical local geometry C2al (directional defect).** The geometric inequality has the wrong direction — the correct form is $g_R/d_m \leq T_m^\beta/\beta_0$ (lower bound on d/g , not upper).
- **Cluster selection via Δ -projection.** Simple δ_N -projection is not a robust cluster selector: at $\lambda = 5, N = 55$ `best_zero=k=16` while the other two select $k = 17$.
- **Log-concavity / detailed balance (C2cc H1/H2/H4).** The hypotheses that β_n or $|b_n|$ are log-concave (H1), that the spectral residue chain has a detailed-balance structure (H2), or that a Mertens-anchor applies (H4) were tested at $\lambda = 3, N = 30, \text{dps} = 50$ and falsified. Structural cause: arithmetic (prime-driven) oscillation.
- **Direct Frontier-Dominance v2.1 (Killer 0, 2026-05-14).** The v2.1 argument used a common Rayleigh test vector to prove $\lambda_{\min}(D_{tot}) < 0$ and then identified $\Delta_{low} = \lambda_{\min}(D_{tot})$ with $\Delta = \lambda_1^+ - \lambda_1^-$. The reviewer counterexample $A = \text{diag}(0, 10), B = \text{diag}(5, -5)$ shows this does *not* follow. Even dominance for asymptotic λ remains conditional on a separate odd-sector lower bound. (This item is Even-sector-specific and documented in detail in Part II [7]; it is listed here because the falsification also restructured the comparative analysis of Branches B and Z.)
- **Euler-diagonal determinant (Z1, closed by Schatten trichotomy, 2026-06-09).** Realising the counting object as a regularised determinant over the Euler diagonal fails structurally: on $\text{Re } s = 1/2$ the diagonal lies in $\mathcal{S}_3 \setminus \mathcal{S}_2$ ($\det_2$ diverges via Mertens), while $\det_3$ loses the arithmetic $k = 1, 2$ kernel. Only a different singular-value class (Ruelle-transfer type) remains unexplored.
- **Contour counting inside Ω_E alone (F3, 2026-06-09).** Closing the zero count by contours lying entirely within the region of established convergence fails by a convexity lemma; any gain re-imports positivity/tightness a priori.
- **Constant- C packet Rouché / infinite-comb reference (2026-06-10/11).** On packet contours, Rouché domination against a constant multiple of Ξ fails in the $N \rightarrow \infty$ limit at all tested c (margin plateau 1.541 at $c = 9$; rising through ≈ 0.95 at $c = 14$); completing the comb to its sinc closed form makes the reference strictly worse (exponential-type clash). The z -dependent normalisation question is settled in Section 3.4.
- **Root-pairing bulk models (B_K , PolyB, 2026-06-09/10).** Modelling the bulk factor by pairing outer secular roots against outer zeros is inert (the roots hit the zeros below measurement floor, $B_K \equiv 1$); the bare local root polynomial against

$C\Xi$ fails by two orders of magnitude (margin 35). The bulk is carried exclusively by the compensation polynomial and the Ξ envelope.

Remark 7.1 (CCM-pattern: cancellation, not magnitude). The CCM-side dead routes share a different pattern from the spectral side: each tries to bound a quantity by its individual-term magnitude, and each is defeated by large internal cancellation. The living routes explicitly factor the cancellation through projector identities or scalar secular equations, never through simple norm domination.

8 Methodical Dead Ends

A small number of methodical routes were tested on the spectral side and ruled out on structural grounds.

- **Dobrushin–Zegarlinski uniqueness $\rightarrow$ RH-CAP.** Dobrushin–Zegarlinski uniqueness works only on lattices with localised Gibbs measures. RH has neither; the absence of multiplicative independence enforces a non-local prime-coupling structure that breaks the Dobrushin contraction setting.
- **Sturm node-counting (naive).** Classical Sturm–Liouville node-counting for the eigenfunctions of QW_N fails because the operator is finite-dimensional and the even/odd spectral gap is sub-exponentially small. Node-counting provides no information at the relevant precision scale.
- **Naive Abel summation for far-tail.** Applying Abel summation to the far-tail prime sums $\sum_{p>\lambda} (\log p)^k / p^s$ without controlling the arithmetic oscillation of $\theta(x) - x$ leads to divergent error terms. The PNT transfer lemma (Part II) shows how to do this correctly.
- **Naive superdeterminant form for Möbius- $\mathbb{Z}_2$.** The formal identity $\text{sdet}(I - L_s) = \det(I - L_s^+) / \det(I - L_s^-) = 1/\zeta(s)$ is algebraically correct for finite prime sets but does not converge for the full prime set at $\Re(s) = \frac{1}{2}$ (trace-class obstruction K4 from Part I).
- **Friedrichs + Petersson as Hilbert–Pólya lever.** The plan to use Friedrichs extension of the Petersson inner product to construct a self-adjoint realisation whose spectrum encodes the Riemann zeros fails: the relevant bilinear form is not semi-bounded on the domain needed to capture the critical-line zeros.

9 Summary

Contribution	Status	Location
I-1 Normal-Family Reframe: (ii) $\rightarrow$ (ii-a) + (ii-c)	rigorous	Section 2.2
(ii-c) absorbed into (ii-a) under (Z)	rigorous	Section 2.2
Foundation-Lock: canonical criterion object	rigorous	Section 2.2
Converged empirics: $A_\lambda(0.4) \approx 1.004$ (7 points)	numerical evidence	Table 1
Multiplicative canonical form	empirical (0.012% fit)	Section 2.4
Theorem B-1: $A_\lambda \leq 1.0063$ (conditional)	conditional	Theorem 2.3
c_Ξ Hadamard curvature (RH-neutral)	analytical	Section 4
CCM: three C2ca-inputs identified	conditional reduction	Section 3.2
CCM: effective microcluster gap $g_{\text{eff}} \approx 5\text{--}6$	numerical evidence	Section 3.3
Determinant channel: exact ledger, bare $P \leftrightarrow \Xi$ pairing, rate $\sigma_1(\gamma > T_{\text{hit}})$	numerical evidence	Section 3.4
6 excluded spectral paths + 13 excluded CCM paths	documented	Sections 6 and 7
5 methodical dead ends	documented	Section 8

The spectral sector contributes two independent conditional reductions of RH, each with a well-characterised open surface. Branch B has the smallest open surface (a single hypothesis (ii-a)) and the strongest empirical support ($A_\lambda \leq 1.0063$ conditional on uniform log-curvature control). The cross-route synthesis across all three branches is developed in Part IV [6]; the condensed atlas and outlook in Part V [5].

Acknowledgments. Computations were performed on a Mac Studio (Apple M1 Max, 32 GB). The interval arithmetic library mpmath was developed by Fredrik Johansson.

AI Disclosure

This manuscript was developed in extensive collaboration with AI language models (Claude, Anthropic; Copilot, Microsoft/GitHub; Gemini, Google DeepMind). AI contributed to conceptual exploration, analytical work, literature review, writing, and critical review. The author, Lukas Geiger, conceived the research direction, provided domain expertise, and exercised final editorial authority. The author assumes full and sole scholarly responsibility for all content, including any errors.

References

- [1] Alain Connes. The Riemann hypothesis: Past, present and a letter through time. *arXiv preprint*, 2026. Survey on the Riemann Hypothesis: past, present, and a letter to Riemann using only pre-1860 mathematics.
- [2] Alain Connes, Caterina Consani, and Henri Moscovici. Zeta zeros and prolate wave operators. *Annals of Functional Analysis*, 15(87), 2024. Jacobi matrix representation and prolate–zeta connection.
- [3] Alain Connes and Walter D. van Suijlekom. Quadratic forms, real zeros and echoes of the spectral action. *Communications in Mathematical Physics*, 406(12), 2025. Quadratic forms, real zeros, and connections to the spectral action.

- [4] Lukas Geiger. A conditional reduction of the Riemann hypothesis to even dominance of the Weil quadratic form. Zenodo, 2026. Standalone extract from the Landscape series; v1.9 record DOI 10.5281/zenodo.20479145, concept DOI 10.5281/zenodo.19764771.
- [5] Lukas Geiger. The Riemann hypothesis: Conclusio. Preprint, 2026. Part V of a five-part series (in preparation).
- [6] Lukas Geiger. The Riemann hypothesis: Cross-route synthesis. Preprint, 2026. Part IV of a five-part series (in preparation).
- [7] Lukas Geiger. The Riemann hypothesis: Even dominance of the Weil quadratic form. Preprint, 2026. Part II of a five-part series.
- [8] Lukas Geiger. The Riemann hypothesis: Foundations, obstructions, and reorientation. Preprint, 2026. Part I of a five-part series.
- [9] Lukas Geiger. The spectral zookeeper: A conditional reduction toward the Riemann hypothesis via spectral microcluster closure in the Connes–Consani–Moscovici framework. Zenodo, 2026. Companion CCM-route paper; v1.5 record DOI 10.5281/zenodo.20479223, concept DOI 10.5281/zenodo.19673126.
- [10] Hugh L. Montgomery. The pair correlation of zeros of the zeta function. *Proceedings of Symposia in Pure Mathematics*, 24:181–193, 1973.
- [11] Edward C. Titchmarsh. *The Theory of the Riemann Zeta-Function*. Oxford University Press, Oxford, 2nd edition, 1986.